



**ReDI School of
Digital Integration**

Final Project Kick-Off

Data Analytics Online

We use tech to connect human potential and
opportunity with dignity & humility

Agenda



- Welcome & Icebreaker: Quick check-in question to engage participants.
- Introduce the final project: Explain the requirements, deliverables, and timeline for the Final Project
- Project ideas and examples
- Determine teams and themes
- Presentation Guidelines
- Check-out & Closing: Final check-out question to reflect on the session and close.



Project

WHY DO THE PROJECT?

- Hands-On Learning:
 - Apply your skills and knowledge to a real project.
- Identify & Improve:
 - Discover what you've mastered and where to grow — ask teachers for guidance!
- Showcase Your Growth:
 - Present your achievements and progress throughout the course.
- Practice Presenting:
 - Practice public speaking in a safe, supportive space—an essential skill for the workplace!

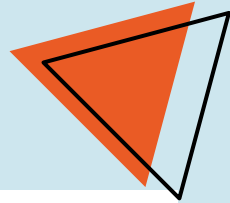
PROJECT REQUIREMENTS



- Work individually or in small groups (up to 3 people)
- Choose your own project topic with teacher guidance
- Submit your project before the Online Class Demo Day to the respective assignment in Google Classroom
- Each project must include:
 - Data cleaning and exploration (with Python and pandas or SQL)
 - Visualizations (with Matplotlib, Seaborn, or Plotly)
 - A well-structured Jupyter Notebook
 - A final presentation showing process, findings, and recommendations

ONLINE DEMO DAY Presentation Requirements

- Length: Max. 5 minutes
- Mode: Slides or Live Demonstration
- Proposed Structure:
 - Intro – What's the project, and what problem does it solve?
 - Demo – Show the live website
 - Challenges & Learning – What went well? What was hard? What did you learn?



Potential Project Ideas



Students will work with provided datasets or find their own, depending on interest and scope.

Project themes may include:

- Gender representation in media or tech
- Environmental data or climate change
- Education and learning outcomes
- City or country-level statistics
- Any dataset that offers meaningful insights and is manageable within the timeline

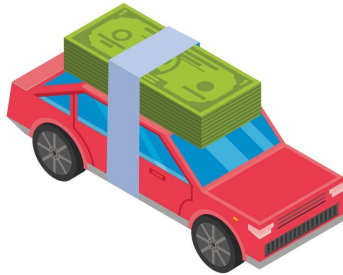


Some Projects from Previous Semesters

Ali



Car Sales and Refunds Analytics



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Introduction (Practical Business Scenario)

You work in a car trade company:

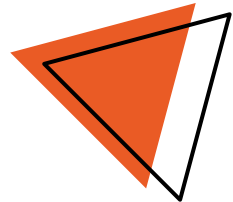
Over the past year, the claims department has seen its expenses going through the roof.

To mitigate the problem, the Director of Operations wants to understand how he can reduce his budget by investigating cars that are causing refunds.

[Source](#)

About the Data

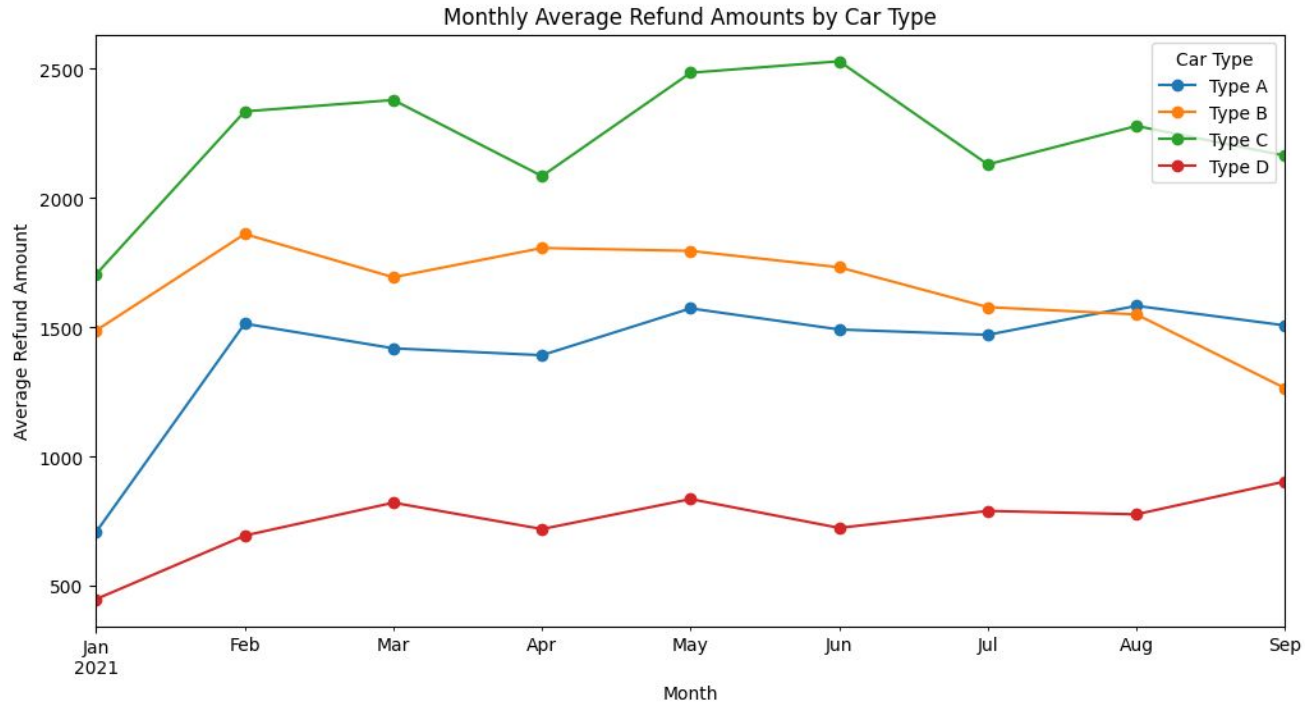
Table	Description	Key Columns
car_sales	Sales information of the cars sold.	Merchant_id, Country, Car_id, Selling_date, Payment_date, Sell_price (EUR), Transport_to_merchant (Boolean), Transport_date
car_details	Car characteristics (including boolean and reference fields).	has_tuning, has_airbags, has_alarm_system, ac_type, fuel_type, gear_type, etc.
car_claims	After-sale claims and their status.	Full_refunds, Partial_refunds, Voucher_amount (EUR), Claim_status



Basic Summary of the Data:

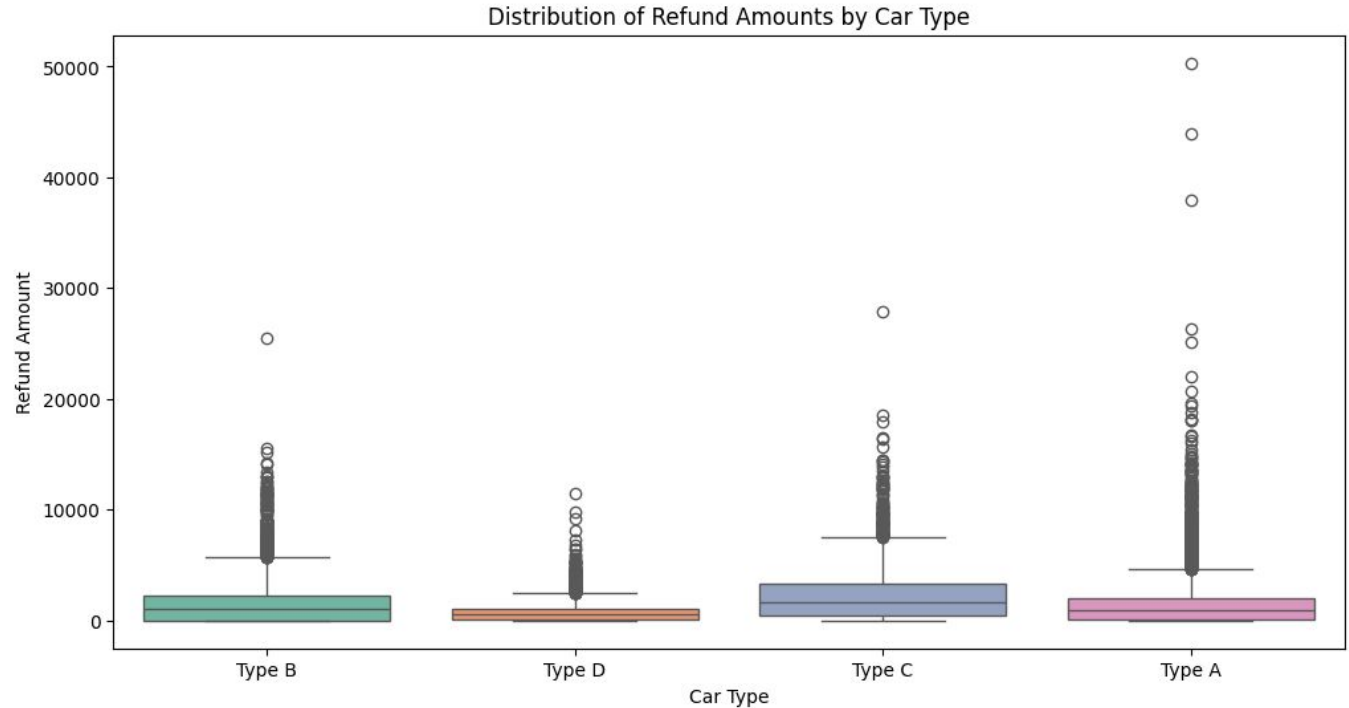
Car Type	Partial Refunds (%)	Full Refunds (%)	Voucher Amount (Mean)	Voucher Amount (Sum)
Type A	76.99	10.63	1488.65	12,516,598.67
Type B	74.66	10.81	1697.92	6,672,831.73
Type C	78.49	9.51	2316.06	5,502,962.87
Type D	76.73	10.98	772.59	3,327,570.71

Comparing the Car types



Distribution of refund amounts

[My Collab](#)



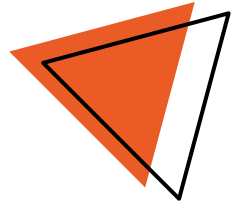
Conclusion & Further analysis

Achievements

- Broke car features into car type by using K mean clustering
- Learned about relationship between car_types and refunds

Next Steps

- Deep dives into car features on Car_type A
- Cross frequency analysis





Diana

Memory Performance

An exploratory behavioral data analysis

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Introduction

What is memory? Is there one kind, or different kinds? And how does our brain remember?

In 2002, Tulving proposes episodic memory (the memory specialized in previous episodes of our lives) as the tool humans use to travel in time.

The data presented was gathered from an experiment developed by Dr. Loris Naspi from CabezaLab as a project to study **false memories**.

The full data sets includes behavioral and neural data, for today I focus on the first exploratory analysis of the behavioral results.

Process

- Start Jupyter notebook*
- Loading data as a dict of multiple data frames
- One dict for each phase (study & test)
- Cleaning for extra symbols, blank spaces, NA's
- Removing subjects
- Defining new df's with columns of interest
- Checking for accuracy and identifying outliers
- Start a memory dataframe per condition to check word and image memory, calculate proportions and z-scores and visualize it.

```
### 3. Extracting encoding (study) data ###
study_files = [f for f in os.listdir(raw_study_results) if f.endswith('.txt')]
study_res_df = {} #Initialize dictionary to hold the dataframes

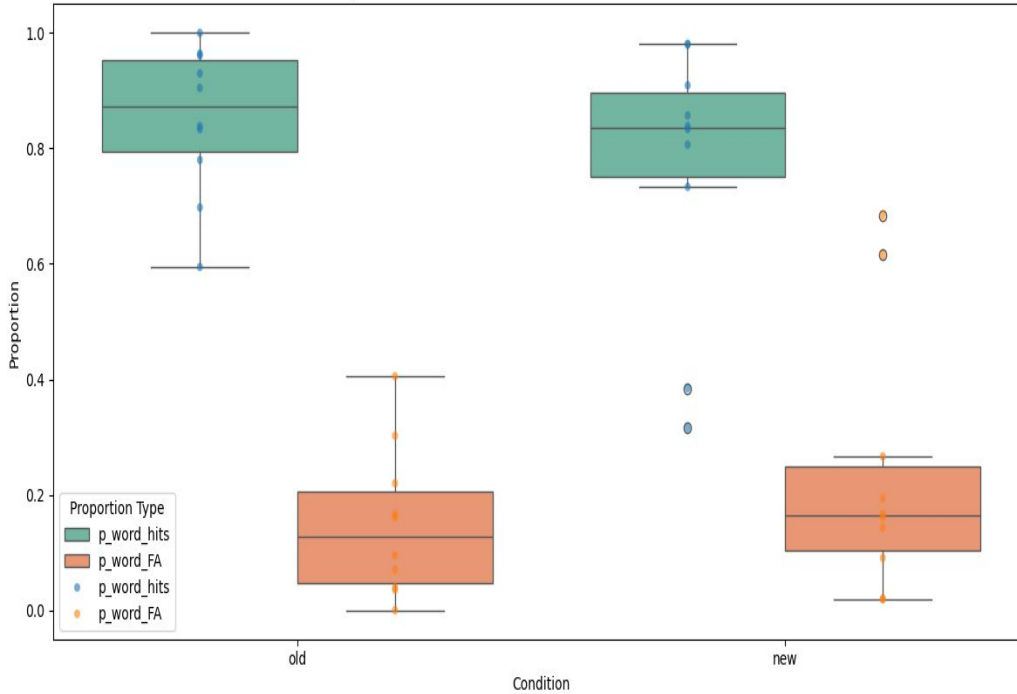
for file in study_files:
    file_path = os.path.join(raw_study_results, file)
    try:
        df = pd.read_csv(file_path, sep='\t')
        subject_id = file.split('_')[-1].split('.')[0]
        study_res_df[f'sub_{subject_id}'] = df
    except Exception as e:
        print(f"Error loading {file_path}: {e}")
```

	participantID	phaselabel	indicator	trial	run	response_word
0	2	retrieval	1	1	1	1.0
1	2	retrieval	2	2	1	4.0
2	2	retrieval	3	3	1	1.0
4	2	retrieval	5	5	1	1.0
5	2	retrieval	6	6	1	1.0
..	...	...	...	...	...	...
234	2	retrieval	235	35	6	4.0
236	2	retrieval	237	37	6	1.0
237	2	retrieval	238	38	6	1.0
238	2	retrieval	239	39	6	1.0
239	2	retrieval	240	40	6	1.0

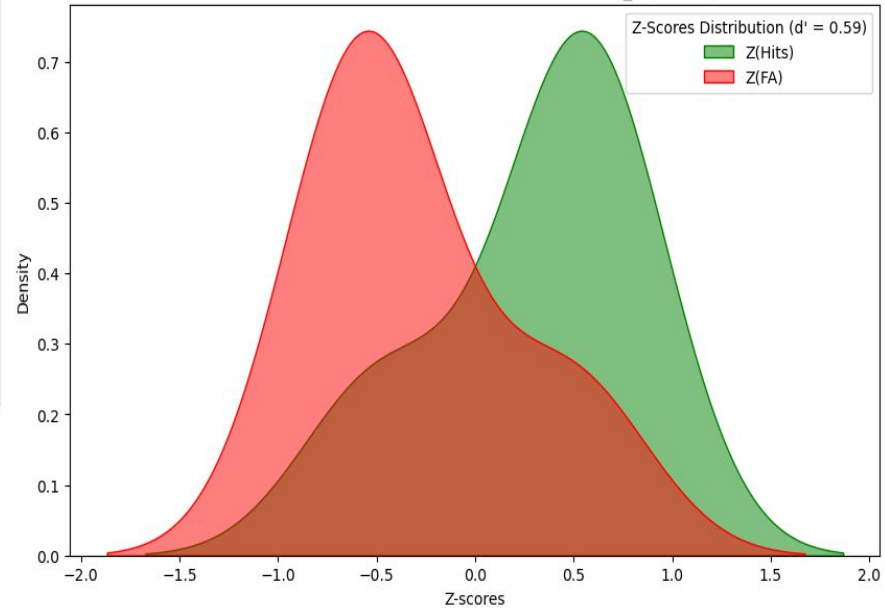


First results

Proportions of Hits and False Alarms of Words Recollection

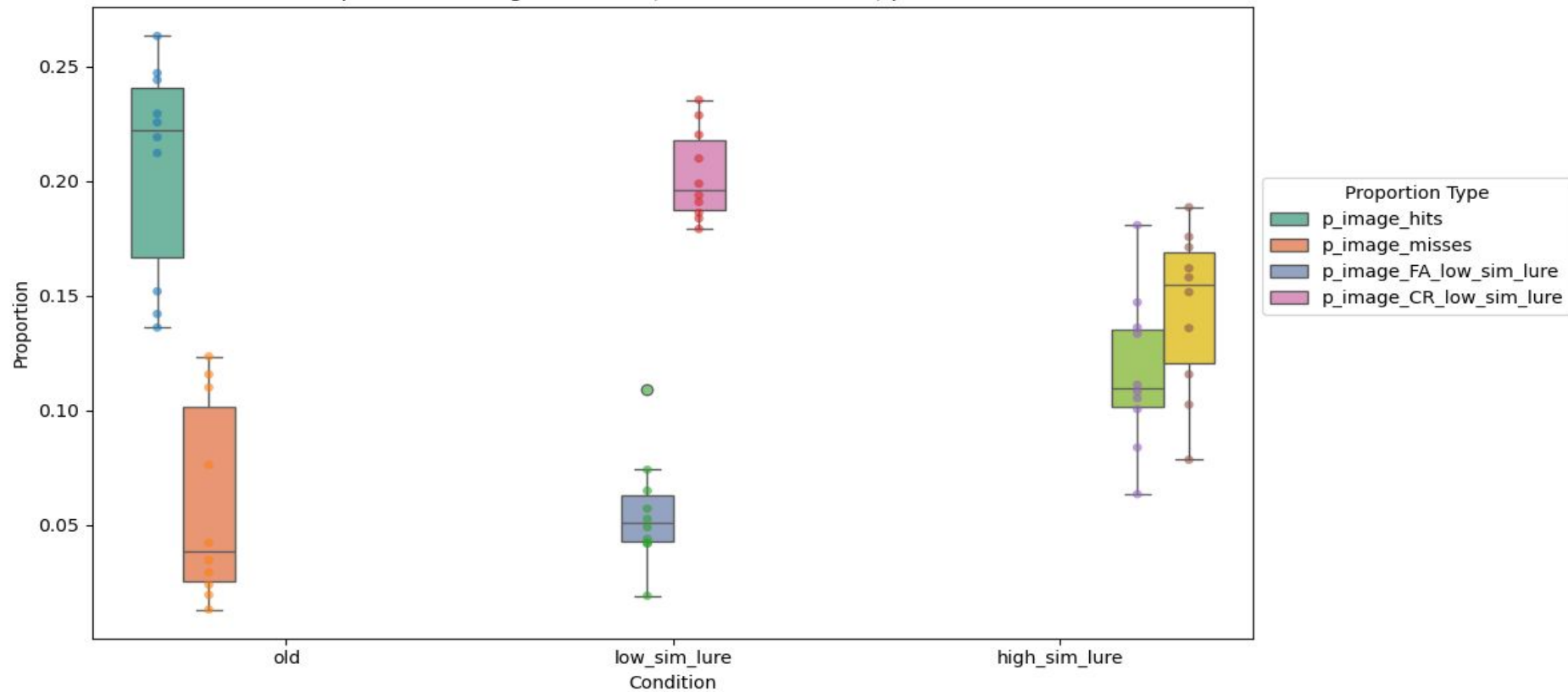


Distributions of Hits and False Alarms for sub_5 (outlier)



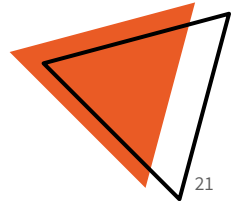
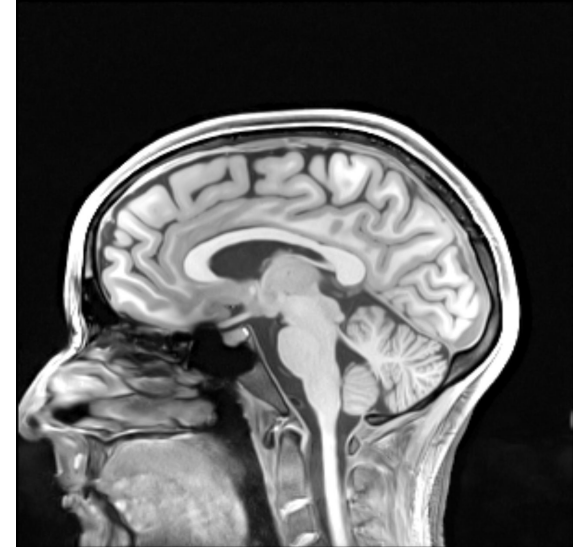
First results

Proportions of Image Retrieval (Hits, Misses, FA, CR) per Condition



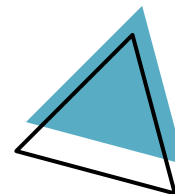
Conclusions & next steps

- Main learnings were how to use python to clean and manage data from several data frames.
- Some issues I encountered were on how to summarize everything into one single general data frame, but it worked out.
- There does seem to be a memory effect! Meaning there looks to be significant difference across the conditions.
- Next steps are: running statistical tests to prove this significance, start fMRI analysis on subjects who pass the behavioral analysis, and write lots!!





Let's Determine Teams and Themes





Project Presentation Guidelines

PRESENTATION STRUCTURE

- Introduction (1 minute):
 - Briefly describe the project and its importance.
 - Explain the need it addresses and the problem it solves.
 - Identify the audience it benefits.
- Core of the Presentation:
 - Focus on the additional research conducted by the team.
 - Ensure classmates learn something new.
- Demo:
 - Include a practical demonstration, case study, or explanation of how the project applies in the real world.
 - Provide specific examples for better understanding.
- Short Conclusions (1 minute):
 - Reflect on how the project could be expanded with more time.
 - Share challenges faced as a team and how they were resolved.
 - Highlight lessons learned throughout the project process.



Useful tips for any presentation - KISS

- Keep it Short and Simple
 - Keep it structured
 - Rehearse and timebox yourself
 - Know your audience

The 6x6 rule - for powerpoint

The 6x6 rule - each slide should:

- Contain no more than **6 words per line**.
- Have no more than **6 lines per slide**.

Additional tips:

- Use visuals (like images, icons, and charts) to spice up your presentation and tell a story.
- Keep font size large enough to be easily readable (at least 14-20 point font size)
- Bullet points (don't read off your slides). No chunky paragraphs of text.

Story format

- Engage your audience with memorable examples
- Set the scene, characters, the task at hand
- There should be - Start, Middle, End
 - Story Arc (intro -> development -> key insight -> resolution -> wrap-up)
- Introduce some drama and tie it up in the conclusion
- Draw from personal stories and experiences. Add some humor/empathy to make a connection with the audience.



Let's form groups! You are free to work on the final project in your current study groups. Group Forming Sheet is linked in the course sheet.

